Business Analysis Report: Customer Support Operations & Performance Analysis

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Business request: The Support Manager and the Operations Head want to know **why customer support tickets are late and which issues to fix first**, using ticket data from January to August 2026.

About this project. This is a portfolio project. The data is **synthetic** (created for practice) and the stakeholders are fictional roles. The ticket types are modelled on the subscription, billing and account requests I handled in my previous role.

1. Background

A customer support team handles requests such as billing issues, subscription cancellations, login problems, account updates, delivery or service complaints and technical faults. Customers are waiting too long for some types of request, and management has no clear view of what is causing the delays. Ticket data exists only as a raw export with duplicates, inconsistent category names and impossible dates. This project cleans the data, analyses it, shows the results in a dashboard and recommends what to fix first.

2. Objectives

- Measure ticket volume, resolution rate and resolution time overall and by category and priority.
- Find which categories and priorities miss their deadlines (SLA) most often.
- Test whether resolution time affects customer satisfaction.
- Recommend what management should fix first.

3. Stakeholders

Role	What they want to know
Operations Head	Where are we missing deadlines, and what should we fix first?
Support Manager	Which ticket types cause the most complaints? What percentage of tickets is resolved?
Team Leads	Which priorities are slowest? Do agents or teams differ?

4. Assumptions

1. The data is synthetic and covers 1 January to 31 August 2026 (4,960 tickets after cleaning).
2. SLA targets (promised resolution times) are: Critical 4 hours, High 24 hours, Medium 48 hours, Low 72 hours.
3. Resolution time is the time from the created date to the resolved date, in hours.
4. Open and Pending tickets (648) have no resolution time, so they are left out of time and SLA measures.
5. Customer rating (CSAT) is a score from 1 to 5. Tickets without a rating are left out of rating averages.
6. Duplicate tickets and tickets with impossible dates are removed before analysis.
7. The 15 agents are grouped into three teams (Alpha, Beta, Gamma). This grouping was created for this project to practise a SQL JOIN.

5. Questions to ask the stakeholders

These are the questions I would ask at the start of the project. The analysis answers most of them.

1. How do you review ticket performance today, and how long does it take?
2. Why do you need this analysis now (late tickets, unhappy customers, cost)?

3. Who will use the results: the support manager, team leads, the operations head?
4. Which measures matter most (resolution rate, resolution time, SLA breach, customer rating)?
5. What are the SLA targets for each priority?
6. When is a ticket "resolved", and should Open and Pending tickets be counted?
7. Which breakdowns do you need: category, priority, channel, month, agent or team?
8. How is the customer rating collected, and what should we do with tickets that have no rating?
9. How far back should the analysis go, and how often will it be repeated?
10. Are there known data problems, such as duplicates, wrong dates or inconsistent names?
11. Do you need a one-time report or a dashboard that managers can filter?
12. Which decisions will you make from the results, and what would success look like?

6. Approach

Step	Tool	What I did	Why
1. Clean the data	Excel	Removed duplicates, standardised category names, removed impossible dates, added columns for resolution hours, month, SLA target and SLA met, built 3 PivotTables, logged every issue.	Wrong data gives wrong answers. The raw data was kept unchanged in a separate sheet.
2. Ask the questions	SQL (MySQL)	Loaded 4,960 tickets and wrote 7 queries (GROUP BY, CASE WHEN, AVG, COUNT, one JOIN). Each query answers one business question.	SQL answers questions on large data in a repeatable way.
3. Show the answers	Power BI	Built a 2-page dashboard with KPI cards, charts and filter boxes for category and priority.	A manager can see the answer at a glance and filter it.
4. Check the numbers	Excel, SQL, Power BI	Compared the same figures across the three tools (for example 4,960 tickets and 86.9% resolution rate).	When three tools agree, the result can be trusted.
5. Recommend	This report	Turned the findings into five recommendations.	Numbers only help if they lead to action.

7. KPI definitions and business rules

KPI	Definition
Total Tickets	Count of all unique tickets after cleaning.
Resolved Tickets	Tickets with status Resolved.
Resolution Rate %	Resolved tickets divided by total tickets.
Avg Resolution Hrs	Average hours from created date to resolved date (tickets that have a resolution time).
SLA Breach %	Share of resolved tickets that took longer than the SLA target for their priority.
Avg CSAT	Average of the 1 to 5 satisfaction score (tickets that have a rating).

Priority	SLA target	Rule
Critical	4 hours	Resolution above 4 hours counts as a breach.
High	24 hours	Resolution above 24 hours counts as a breach.
Medium	48 hours	Resolution above 48 hours counts as a breach.
Low	72 hours	Resolution above 72 hours counts as a breach.

8. Data quality findings

The raw file had 5,050 rows. After cleaning, 4,960 tickets remained.

Issue found	Count	Action taken
Duplicate tickets	50	Removed
Inconsistent category names (capital letters, extra spaces)	about 120	Standardised with PROPER and TRIM. Six clean categories.
Resolved date earlier than created date (impossible)	40	Removed from the dataset
Ticket marked Resolved but no resolved date	58	Kept, but left out of resolution time measures
Resolved ticket with no satisfaction score	about 300 (297 in the final data)	Kept, but left out of rating averages

9. Findings

All numbers come from the SQL queries (Appendix C) and match the dashboard (Appendices A and B).

Overall results

Measure	Result
Total tickets (cleaned)	4,960
Resolution rate	86.9% (4,312 resolved)
Average resolution time	33.2 hours
SLA breach rate	about 28% of resolved tickets
Average customer rating (1 to 5)	3.68

Finding 1: Billing issues are the biggest problem

Billing is 28% of all tickets, but 49.3% of billing tickets miss their deadline. Billing accounts for 593 of the 1,211 missed deadlines, which is about 49% of all of them. Technical Fault has the second-highest breach rate (48.8%), although it is only 7.8% of tickets. Billing and Technical Fault together cause about 62% of all missed deadlines.

Category	Tickets	Share of tickets	Resolved (with time)	Missed deadline	Missed %	Share of all misses
Billing Issue	1,394	28.1%	1,204	593	49.3%	49.0%
Technical Fault	385	7.8%	334	163	48.8%	13.5%
Delivery/Service Complaint	580	11.7%	491	176	35.8%	14.5%
Subscription Cancellation	1,014	20.4%	874	176	20.1%	14.5%
Account Update	696	14.0%	585	55	9.4%	4.5%
Login Problem	891	18.0%	766	48	6.3%	4.0%

Finding 2: Critical tickets are far too slow

Critical tickets take 11.8 hours on average against a 4-hour target, nearly three times the limit. High tickets average 23.0 hours against 24, so many of them are late. Medium and Low tickets are within target on average, although averages can hide individual late tickets.

Priority	Average hours	SLA target (hours)
Critical	11.8	4
High	23.0	24
Medium	34.3	48
Low	47.2	72

Finding 3: Slow tickets lead to unhappy customers

The longer a ticket takes, the lower the rating. Tickets solved in under 24 hours average 4.37, and tickets taking over 48 hours average 2.17. The 797 tickets in the slowest group are about 20% of all rated tickets.

Time to resolve	Tickets rated	Average rating
Under 24 hours	1,819	4.37
24 to 48 hours	1,345	3.63
Over 48 hours	797	2.17

Finding 4: The cause is ticket type and process, not people or channels

Across the 15 agents, the average resolution time ranges only from 30.2 to 36.1 hours and ratings from 3.52 to 3.76. The three teams are also close (32.5 to 33.9 hours, ratings 3.65 to 3.71), and average ratings by channel are similar (3.63 to 3.77). So differences between agents, teams and channels do not explain the delays.

10. Recommendations

Expected benefits are illustrative scenarios based on this data, not guarantees.

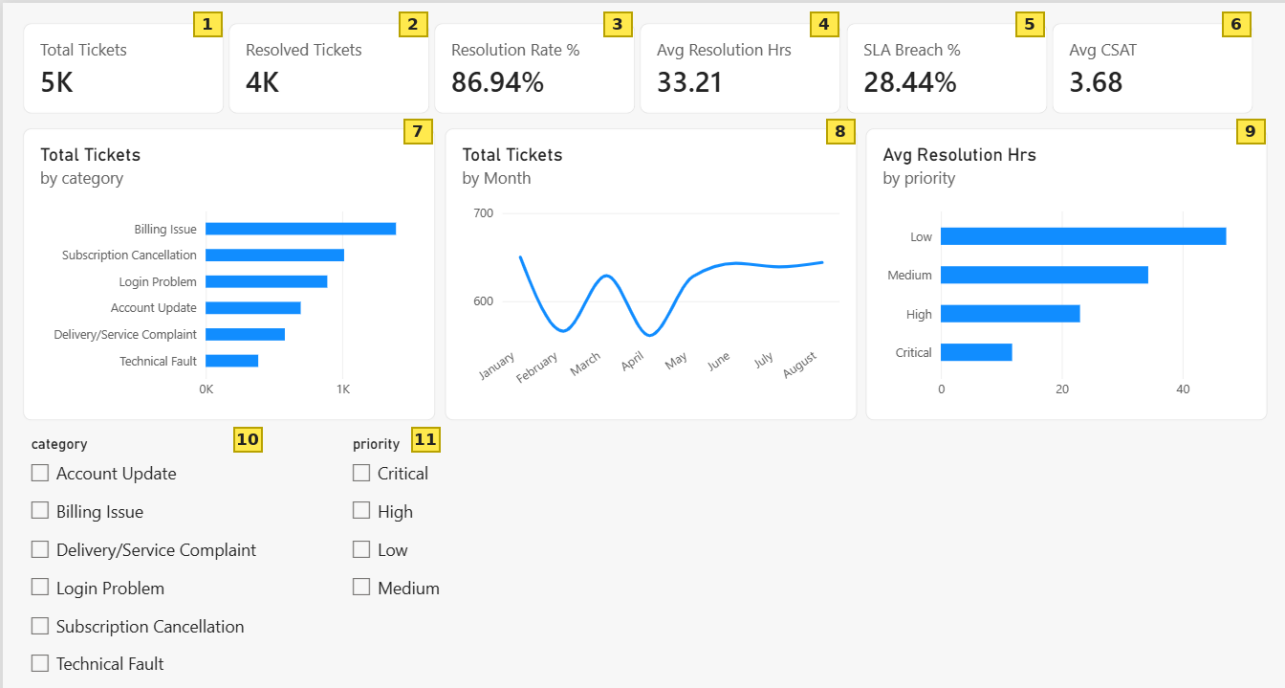
#	Finding	Recommendation	Expected benefit
1	Billing causes about 49% of all missed deadlines (49.3% breach rate).	Add a billing self-service FAQ and templated replies. Train a small billing specialist group.	If the billing breach rate fell from 49% to 35%, the overall breach rate would fall from about 28% to about 24%.
2	Critical tickets average 11.8 hours against a 4-hour target.	Create a fast-track queue for Critical tickets with an alert after 2 hours.	Fewer missed deadlines on the highest-impact tickets.
3	Ratings fall from 4.37 to 2.17 after 48 hours.	Escalate any ticket open for 36 hours to a team lead.	Fewer tickets reach the lowest-rated group.
4	Technical Fault tickets breach 48.8% of the time despite low volume (385).	Review the hand-off to technical teams and agree clearer response times.	Targets the second-highest breach rate with a small number of tickets.
5	Agents, teams and channels perform similarly.	Focus on ticket type and process, not individual performance reviews.	Effort goes to the real cause.

How the 24% in recommendation 1 is calculated: a fall from 49.3% to 35% on 1,204 billing tickets is about 172 fewer late tickets. Late tickets would drop from 1,211 to 1,039 out of 4,254 resolved tickets, which is 24.4%.

11. Limitations and next steps

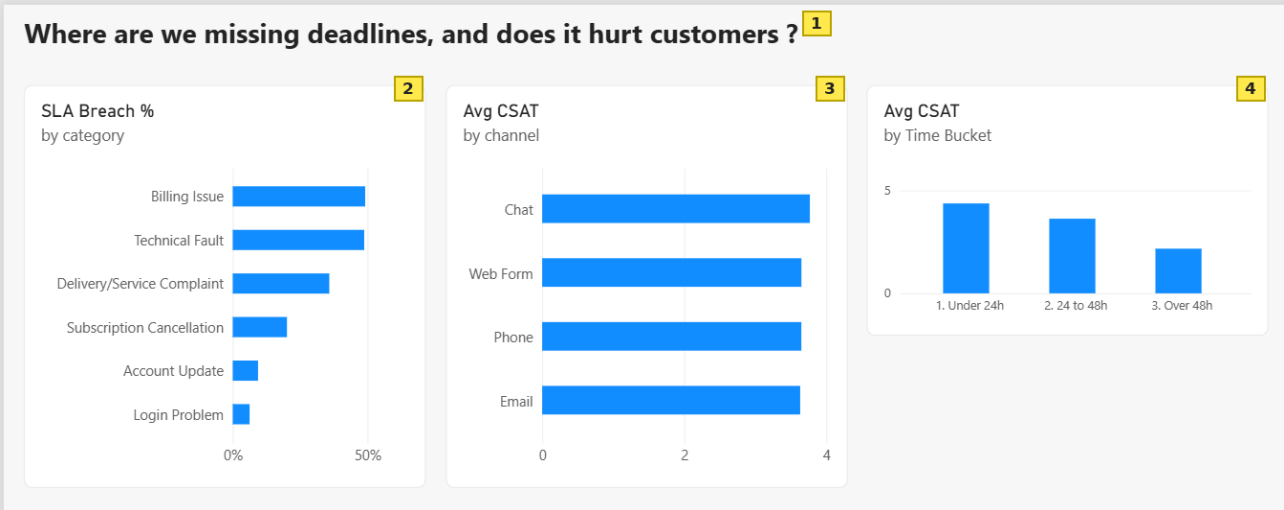
- The data is synthetic, so the patterns are illustrative. Real data would need the same checks before any decision.
- The link between slow tickets and low ratings shows an association, not proof of cause.
- Open and Pending tickets have no resolution time, so time-based results describe resolved tickets only.
- The SLA targets are assumed and should be confirmed with the stakeholders.
- In this version of the dashboard, the Total Tickets and Resolved Tickets cards display rounded values (5K and 4K). The exact values are 4,960 and 4,312.
- **Next steps:** use real data, confirm the SLA targets, add an agent or team view, and add data labels to the charts.

Appendix A: Dashboard Page 1 (overview)



ID	Item	Type	What it shows	Notes
1	Total Tickets	Number card	Number of tickets	Displays 5K (rounded). Exact value 4,960.
2	Resolved Tickets	Number card	Tickets with status Resolved	Displays 4K (rounded). Exact value 4,312.
3	Resolution Rate %	Number card	Resolved tickets divided by total tickets	86.94%
4	Avg Resolution Hrs	Number card	Average hours from created to resolved	33.21 hours
5	SLA Breach %	Number card	Share of resolved tickets later than their SLA target	28.44%
6	Avg CSAT	Number card	Average customer rating, 1 to 5	3.68
7	Total Tickets by category	Bar chart	Tickets per category, highest first	Billing Issue is highest (1,394).
8	Total Tickets by month	Line chart	Tickets per month, January to August	Between 561 and 650 tickets a month.
9	Avg Resolution Hrs by priority	Bar chart	Average hours for Critical, High, Medium and Low	Low is longest (47.2), Critical shortest (11.8).
10	category	Filter box	Choose one or more of the six categories	Updates all Page 1 cards and charts. Click again to clear.
11	priority	Filter box	Choose one or more priorities	Updates all Page 1 cards and charts. Click again to clear.

Appendix B: Dashboard Page 2 (SLA and satisfaction)



ID	Item	Type	What it shows	Notes
1	Page title	Text	The question this page answers	"Where are we missing deadlines, and does it hurt customers?"
2	SLA Breach % by category	Bar chart	Share of resolved tickets later than their SLA, per category	Billing Issue 49.3% and Technical Fault 48.8% are highest. Login Problem is lowest (6.3%).
3	Avg CSAT by channel	Bar chart	Average rating for Chat, Web Form, Phone and Email	Ratings are similar (3.63 to 3.77), so channel is not the cause.
4	Avg CSAT by Time Bucket	Column chart	Average rating for Under 24h, 24 to 48h and Over 48h	4.37, 3.63 and 2.17. Tickets without a resolution time are not shown.

Page 2 has no filter boxes in this version.

Appendix C: SQL queries

Database: support_db, table: tickets (4,960 rows). The full script is support_analysis_queries.sql and the result screenshots are in the sql folder of the project.

Q1. Which categories have the most tickets?

```
SELECT category, COUNT(*) AS total_tickets
FROM tickets
GROUP BY category
ORDER BY total_tickets DESC;
```

Q2. Which priorities take the longest to resolve?

```
SELECT priority, ROUND(AVG(resolution_hours), 1) AS avg_hours
FROM tickets
WHERE resolution_hours IS NOT NULL
GROUP BY priority
ORDER BY avg_hours;
```

Q3. What percentage of tickets are resolved?

```
SELECT COUNT(*) AS total_tickets,
       SUM(CASE WHEN status = 'Resolved' THEN 1 ELSE 0 END) AS resolved_tickets,
       ROUND(100 * SUM(CASE WHEN status = 'Resolved' THEN 1 ELSE 0 END)
             / COUNT(*), 1) AS resolution_rate_pct
FROM tickets;
```

Q4. Which categories miss their SLA deadline most often?

```
SELECT category,
       COUNT(*) AS resolved_tickets,
       SUM(CASE WHEN sla_met = 'No' THEN 1 ELSE 0 END) AS missed_deadline,
       ROUND(100 * SUM(CASE WHEN sla_met = 'No' THEN 1 ELSE 0 END) / COUNT(*), 1) AS missed_pct
FROM tickets
WHERE sla_met IS NOT NULL
GROUP BY category
ORDER BY missed_pct DESC;
```

Q5. Does resolution time affect customer satisfaction?

```
SELECT CASE WHEN resolution_hours < 24 THEN '1. Under 24h'
            WHEN resolution_hours <= 48 THEN '2. 24 to 48h'
            ELSE '3. Over 48h' END AS time_bucket,
       COUNT(*) AS tickets,
       ROUND(AVG(satisfaction_score), 2) AS avg_rating
FROM tickets
WHERE resolution_hours IS NOT NULL AND satisfaction_score IS NOT NULL
GROUP BY time_bucket
ORDER BY time_bucket;
```

Q6. How do agents compare on speed and rating?

```
SELECT agent_id,
       COUNT(*) AS tickets,
       ROUND(AVG(resolution_hours), 1) AS avg_hours,
       ROUND(AVG(satisfaction_score), 2) AS avg_rating
FROM tickets
WHERE resolution_hours IS NOT NULL
GROUP BY agent_id
ORDER BY avg_hours;
```

Q7. How do the teams compare? (uses a JOIN with the agents table)

```
SELECT a.team,
       COUNT(*) AS tickets,
       ROUND(AVG(t.resolution_hours), 1) AS avg_hours,
       ROUND(AVG(t.satisfaction_score), 2) AS avg_rating
FROM tickets t
JOIN agents a ON t.agent_id = a.agent_id
WHERE t.resolution_hours IS NOT NULL
GROUP BY a.team
ORDER BY avg_hours;
```